Hassan Shahzad

Software Engineer | Forward Deployed Engineer

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Summary

Full-stack software engineer specializing in AI-powered SaaS, RAG systems, and production backend architectures. Experienced in owning products end-to-end across forward-deployed and founding engineering roles, from system architecture and third-party integrations to AI workflows, payments, and real-time applications, with a track record of shipping production features that solve complex real-world problems.

Experience

Software Engineer | Forward Deployed Engineer - *Full-time* SEP 2025 - PRESENT

Softpers

- Outersuite: Construction Proposals & Cost Estimation
  - Built a full-stack SaaS platform using **TypeScript**, **React**, **FastAPI**, and **PostgreSQL** for commercial exterior-services workflows, integrating **Salesforce**, aerial imagery, property measurement, cost estimation, bid generation, and subcontractor RFIs.
  - Integrated **7** external systems, including **Salesforce**, **Nearmap**, **Monday**, **Google Drive**, **RSMeans**, and **OpenStreetMap**.
  - Replaced a paid Nearmap AI dependency with a zero-cost geospatial pipeline using OSM building footprints and USGS 3DEP LiDAR for automated building tracing and height extraction, then built interactive takeoff workflows on top of it using **TypeScript**, **Leaflet**, **OpenLayers**, **TanStack Query**, and **Zustand**.
  - Built 4 production Claude-powered features, including an SSE-streaming assistant, PDF-to-JSON parser, AI bid summaries, and batched cost classification.
  - Engineered a deterministic **FastAPI** cost engine using Decimal arithmetic with reconciliation across labor, fees, freight, markup, and tax, and added **Sentry**-based error monitoring for production reliability.
- Vynsite: Automotive Dealership & Customer Engagement
  - Architected a multi-tenant automotive SaaS platform using **NestJS**, **TypeScript**, **PostgreSQL**, and **Prisma** for dealership operations, loyalty, referrals, inventory, and lead management.
  - Designed a domain-driven backend with **RBAC**, **2FA**, audit logging, and secure **multi-tenant access** controls.
  - Built AI-powered customer insights and vehicle recommendations using the **OpenAI API**.
  - Integrated inventory, VIN decoding, fuel economy, and geolocation providers.
- Voyajur: Vacation Rental & Property Management
  - Architected a booking and property management platform using **Next.js**, **Express**, **GraphQL**, **PostgreSQL**, and **Prisma**.
  - Designed a reservation domain covering bookings, payments, messaging, reviews, PMS integrations, and operations.
  - Implemented **JWT authentication**, **RBAC**, and real-time authorization.
  - Built **Stripe** payment processing with webhooks, refunds, and reconciliation.
  - Engineered Redis-backed queues, real-time communication, and channel-manager integrations for **Smily**, **Guesty**, **Hostaway**, and **iCal**.
- Hilo ATOS: Aviation Training & Operations Platform
  - Architected a multi-tenant aviation training and operations platform using **Next.js**, **TypeScript**, and **PostgreSQL**, with a portable backend that runs in both serverless and traditional environments.
  - Integrated **Stripe Connect**, **AWS S3**, **SendGrid**, **Google Calendar**, and FAA data sources.
  - Delivered AI-powered scheduling and document assistance using **Anthropic Claude**.

Founding Software Engineer - *Contract* OCT 2025 - MAY 2026

Folklore

Folklore (formerly Echotree) is an AI-powered memory and legacy platform that turns a person's stories, voice, and personality into an interactive digital Heirloom that loved ones can talk to and learn from.

- Architected and built a production AI platform that converts interview transcripts into voice-enabled conversational personas, serving **20+** active users, supporting **5+** client families, and powering **60+** live AI personas.
- Led full-stack development using **Next.js 15**, **TypeScript**, **Supabase**, and **PostgreSQL**, owning architecture, authentication, conversations, voice experiences, administration workflows, and billing systems.
- Solely architected and developed a standalone **FastAPI-based RAG microservice** using **Qdrant**, **SentenceTransformers**, and **Google Gemini API**, reducing hallucinations by **70%** compared to general-purpose AI responses.
- Implemented real-time conversational and voice experiences using **speech-to-text**, **ElevenLabs TTS**, and **encrypted audio caching (AES-256-GCM)**, reducing voice response latency by **65%** and lowering TTS infrastructure costs by **45%**.
- Integrated **Stripe** subscriptions, billing portal workflows, and a metered token quota system, supporting **$7K+** annual recurring revenue across **4** subscription tiers with automated webhook processing and usage tracking.
- Containerized and deployed AI services using **Docker** and **Railway**, reducing image size from 8.2 GB to 3.7 GB, with health monitoring, structured logging, auto-recovery, and observability through **Sentry** and **PostHog**.
- Built a multi-step persona creation pipeline including transcript processing, AI-generated stress-test questions, retrieval indexing, voice configuration, and activation workflows, reducing persona QA and onboarding time from **6 hours** to **under 90 minutes**.

- Engineered high-performance **RAG inference pipelines** with semantic retrieval, intelligent chunking, cosine similarity, MMR re-ranking, and persona-aware prompt orchestration with **Google Gemini**, achieving **<120ms** retrieval latency, **40%** faster time-to-first-token, and **150+** concurrent requests per instance.

Software Development Engineer - I - *Full-time* FEB 2025 - AUG 2025

Arrivy

Arrivy is a field service coordination platform that streamlines customer communication, crew management, and operational logistics in real time.

- Built and maintained **Python** and **Node** microservices for PDF handling and database tasks, improving modularity and deployment speed.
- Developed and deployed **Google Cloud Functions** and **Pub/Sub** pipelines to enable real-time crew attendance tracking, logging schedules, and work hours with high reliability and a scalable architecture.
- Built an internal authorization tool (**React** + **Flask**) for account access management, improving auditing and traceability.
- Leveraged **Redis** queues to run background tasks asynchronously, implementing caching to cut user-facing latency by over **60%** during PDF processing.
- Implemented dynamic PDF generation and compression pipelines using **puppeteer** and **ghostscript4js** libraries, optimizing the process and reducing PDF sizes by **80%** without loss in quality.
- Designed and implemented a **React** and **Flask** based template system for task creation, optimizing workflows by auto-copying files attached to templates to tasks on selection, accelerating workflows, and reducing manual effort.
- Implemented a secure DB authentication system using sandboxed service accounts to access production data via **SQLAlchemy ORM**.
- Integrated third-party APIs (**HubSpot**, **Workday**, **Easybook**) using a **Node** and **Python** based backend for real-time data sync and workflow automation.

Technical Content Engineer - *Full-time* JUL 2023 - JAN 2025

Educative

- Curated **React** and **Node.js** technical content and hands-on projects covering REST APIs, databases, authentication, and production deployment, improving learner engagement and course completion.
- Contributed to **AI mock interviewers** through prompt engineering, model testing, and feedback optimization across multiple ML models.
- Integrated **Generative AI** and **LLMs** into coding widgets for real-time mentorship, while building Docker-based browser environments for thousands of concurrent users, increasing engagement by **25%**.

Education

FAST National University of Computer and Emerging Sciences AUG 2019 - JUN 2023

B.S., Computer Science

Relevant courses: Web Programming, Software Engineering, Artificial Intelligence, Natural Language Processing, Software Design and Analysis, Software Testing.

Technical Skills

- **Frontend & Full-Stack Development:** TypeScript, JavaScript, React.js, Next.js, Tailwind CSS, TanStack Query, Zustand, Server-Sent Events (SSE), WebSockets, Real-Time Applications
- **Backend & APIs:** Python, FastAPI, Node.js, Express.js, NestJS, RESTful APIs, GraphQL, Apollo GraphQL, Microservices, Service-Oriented Architecture, Authentication & Authorization, JWT, RBAC, 2FA, Middleware, Request Validation, Background Jobs, Webhooks, API Integration
- **AI & Machine Learning:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, AI Agents & Orchestration, Anthropic Claude, Gemini, OpenAI API, Hugging Face, SentenceTransformers, Qdrant, Vector Databases, Embeddings, Semantic Search, Cosine Similarity, Maximum Marginal Relevance (MMR), Intelligent Chunking, AI Document Processing, AI Summarization & Classification
- **Databases & Data Systems:** PostgreSQL, Supabase, MongoDB, Redis, Google Firestore, BigQuery, Prisma, SQLAlchemy ORM, Row-Level Security (RLS), Database Schema Design, Normalization, Indexing & Query Optimization, Transactions, Connection Pooling, Caching, Multi-Tenant Data Isolation
- **Cloud, DevOps & Infrastructure:** Google Cloud Platform (GCP), Cloud Run, Cloud Functions, Pub/Sub, Cloud Storage, Cloud Scheduler, AWS S3, Docker, Railway, Vercel, Artifact Registry, Cloud Build, CI/CD, Sentry, PostHog, Monitoring & Logging
- **Payments & Integrations:** Stripe, Stripe Connect, Subscription Billing, Metered Usage & Quotas, Payment Webhooks, Billing Reconciliation, Salesforce, Workday, Google Drive, Monday, Guesty, Hostaway, and iCal
- **Geospatial & Data Processing:** Leaflet, OpenLayers, OpenStreetMap (OSM), USGS 3DEP LiDAR, Geospatial Data Processing, Building Tracing, Height Extraction, PDF Processing, Puppeteer, Ghostscript, Document Parsing

Projects

- Wordley.ai: Built an AI-powered NLP analyzer using React, FastAPI, Hugging Face, and open-source LLMs, delivering insights on sentiment, tone, emotion, summarization, and entities; gained 120+ users in the first month.
- Safelinks: Built a secure temporary-link microservice using Node.js, Express, Redis, Firestore, and Docker, deployed on Google Cloud Run with token-based access and automatic expiration.
- Recaller: Built an AI knowledge base using React, Flask, vector embeddings, similarity search, and DeepSeek, enabling contextual Q&A and summarization over uploaded documents.
- Respoon: Built a food-sharing platform using React, Flask, PostgreSQL, and Socket.IO, featuring geolocation-based discovery, real-time chat, filtering, and pagination.